

PRINCIPLES OF GEOGRAPHIC INFORMATION SYSTEM SEM: VI

SEM VI: UNIT 4



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1. List the four classifications of analytical functions of GIS. Explain any one in details. (Apr 19)

Ans:

❖ **Classifications of analytical functions of GIS:**

There are many ways to classify the analytical functions of a GIS.

1 Classification, retrieval, and measurement functions:

All functions in this category are performed on a single (vector or raster) data layer, often using the associated attribute data.

- Classification allows the assignment of features to a class on the basis of attribute values or attribute ranges (definition of data patterns). On the basis of reflectance characteristics found in a raster, pixels may be classified as representing different crops, such as potato and maize.
- Retrieval functions allow the selective search of data. We might thus retrieve all agricultural fields where potato is grown.
- Generalization is a function that joins different classes of objects with common characteristics to a higher level (generalized) class.
- Measurement functions allow the calculation of distances, lengths, or areas.

2 Overlay functions:

These belong to the most frequently used functions in a GIS application. They allow the combination of two (or more) spatial data layers comparing them position by position, and treating areas of overlap—and of non-overlap—in distinct ways.

Many GISs support overlays through an algebraic language, expressing an overlay function as a formula in which the data layers are the arguments. In this way, we can find

3 Neighbourhood functions:

Whereas overlays combine features at the same location, neighbourhood functions evaluate the characteristics of an area surrounding a feature's location. A neighbourhood function 'scans' the neighbourhood of the given feature(s), and performs a computation on it.

4 Connectivity functions:

These functions work on the basis of networks, including road networks, water courses in coastal zones, and communication lines in mobile telephony. These networks represent spatial linkages between features.

2. Write a note on automatic classification.

(Apr 19)

Ans:

❖ Automatic classification:

User-controlled classifications require a classification table or user interaction. GIS software can also perform automatic classification, in which a user only specifies the number of classes in the output data set.

The system automatically determines the class break points.

Two main techniques of determining break points are in use.

1. Equal interval technique:

The minimum and maximum values v_{min} and v_{max} of the classification parameter are determined and the (constant) interval size for each category is calculated as $(v_{max} - v_{min})/n$, where n is the number of classes chosen by the user.

This classification is useful in revealing the distribution patterns as it determines the number of features in each category.

2. Equal frequency technique:

This technique is also known as quantile classification.

The objective is to create categories with roughly equal numbers of features per category.

The total number of features is determined first and by the required number of categories, the number of features per category is calculated.

The class break points are then determined by counting off the features in order of classification parameter value.

1	1	1	2	8
4	4	5	4	9
4	3	3	2	10
4	5	6	8	8
4	2	1	1	1

(a) original raster

1	1	1	1	4
2	2	3	2	5
2	2	2	1	5
2	3	3	4	4
2	1	1	1	1

(b) equal interval classification

1	1	1	2	5
3	3	4	3	5
3	2	2	2	5
3	4	4	5	5
3	2	1	1	1

(c) equal frequency classification

original value	new value	# cells
1,2	1	9
3,4	2	8
5,6	3	3
7,8	4	3
9,10	5	2

original value	new value	# cells
1	1	6
2,3	2	5
4	3	6
5,6	4	3
8,9,10	5	5

Figure 6.11: Example of two automatic classification techniques: (a) the original raster with cell values; (b) classification based on equal intervals; (c) classification based on equal frequencies. Below, the respective classification tables, with a tally of the number of cells involved.

3. Explain vector overlay operations using suitable diagram.

(Apr 19)

OR

Explain vector overlay operations using suitable diagram.

(Nov 19)

OR

Write short note on vector overlay operation.

(Apr 23)

Ans:

➤ **Vector overlay operations:**

In the vector domain, overlay is computationally more demanding than in the raster domain. Here we will only discuss overlays from polygon data layers, but we note that most of the ideas also apply to overlay operations with point or line data layers.

The standard overlay operator for two layers of polygons is the polygon intersection operator. It is fundamental, as many other overlay operators proposed in the literature or implemented in systems can be defined in terms of it. The principles are illustrated in Figure.

Two more polygon overlay operators are illustrated in Figure. The first is known as the polygon clipping operator. It takes a polygon data layer and restricts its spatial extent to the generalized outer boundary obtained from all (selected) polygons in a second input layer. Besides this generalized outer boundary, no other polygon boundaries from the second layer play a role in the result.

A second overlay operator is polygon overwrite. The result of this binary operator is defined is a polygon layer with the polygons of the first layer, except where polygons existed in the second layer, as these take priority.

Vector overlays are usually also defined for point or line data layers. Their definition parallels the definitions of operators discussed above. Different GISs use different names for these operators, and one is advised to carefully check the documentation before applying any of these operators.

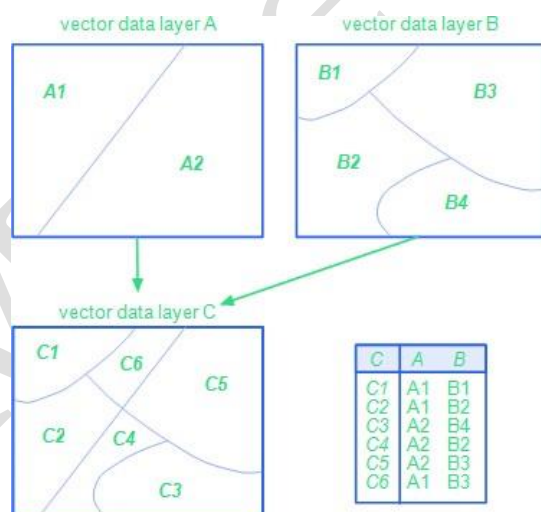


Figure 6.12: The polygon intersect (overlay) operator. Two polygon layers *A* and *B* produce a new polygon layer (with associated attribute table) that contains all intersections of polygons from *A* and *B*. Figure after [8].

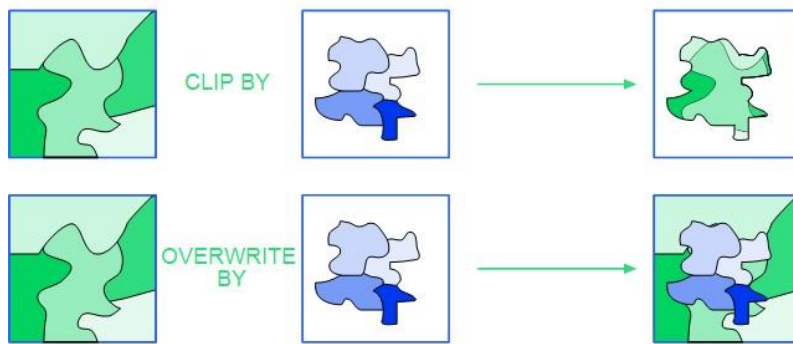


Figure 6.14: Two more polygon overlay operators: (a) polygon clip overlay clips down the left hand polygon layer to the generalized spatial extent of the right hand polygon layer; (b) polygon overwrite overlay overwrites the left hand polygon layer with the polygons of the right hand layer.

4. Explain using example how Raster overlay operation can be performed using decision table? (Apr 19)

Ans:

❖ Raster overlay operation:

Raster overlay operations involve combining multiple raster datasets to derive a new raster output. These operations are essential in GIS for tasks such as land suitability analysis, habitat modeling, and environmental impact assessments. Decision tables can be used to guide the overlay process by defining rules that determine the output value based on the values in the input rasters.

Conditional expressions are powerful tools in cases where multiple criteria must be taken into account.

We could produce the output raster of Figure with a map algebra expression such as:
 Suitability: = CON ((Landuse = "Forest" AND Geology = "Alluvial") OR (Landuse = "Grass" AND Geology = "Shale"), "Suitable", "Unsuitable")

Some GISs accommodate setting up a separate decision table that will guide the raster Overlay process. This extra table carries domain expertise, and dictates which combinations of input raster cell values will produce which output raster cell value. This gives us a raster overlay operator using a decision table, as illustrated in Figure. The GIS will have supporting functions to generate the additional table from the input rasters, and to enter appropriate values in the table.

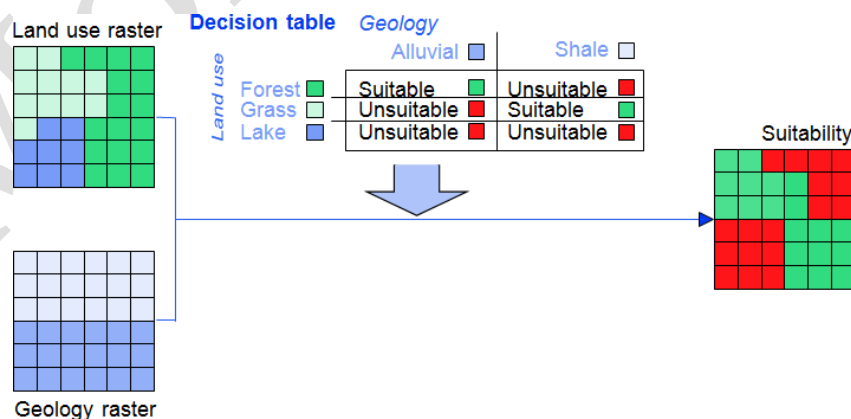


Figure 6.19: The use of a decision table in raster overlay. The overlay is computed in a suitability study, in which land use and geology are important factors. The meaning of values in both input rasters, as well as the output raster can be understood from the decision table.

5. List any five examples where advanced computations on continuous fields are required. (Apr 19)

Ans:

There are numerous examples where more advanced computations on continuous field representations are needed. A short list is provided below.

- **Slope angle calculation:**
The calculation of the slope steepness, expressed as an angle in degrees or percentages, for any or all locations.
- **Slope aspect calculation:**
The calculation of the aspect (or orientation) of the slope in degrees (between 0 and 360 degrees), for any or all locations.
- **Slope convexity/concavity calculation:**
Slope convexity—defined as the change of the slope (negative when the slope is concave and positive when the slope is convex)—can be derived as the second derivative of the field.
- **Slope length calculation:**
With the use of neighbourhood operations, it is possible to calculate for each cell the nearest distance to a watershed boundary (the upslope length) and to the nearest stream (the downslope length). This information is useful for hydrological modelling.
- **Hillshading:**
Is used to portray relief difference and terrain morphology in hilly and mountainous areas. The application of a special filter to a DEM produces hillshading.
- **Three-dimensional map:**
Display with GIS software, three-dimensional views of a DEM can be constructed, in which the location of the viewer, the angle under which s/he is looking, the zoom angle, and the amplification factor of relief exaggeration can be specified. Three-dimensional views can be constructed using only a predefined mesh, covering the surface, or using other rasters (e.g. a hillshading raster) or images (e.g. satellite images) which are draped over the DEM.
- **Determination of change in elevation through time:**
The cut-and-fill volume of soil to be removed or to be brought in to make a site ready for construction can be computed by overlaying the DEM of the site before the work begins with the DEM of the expected modified topography. It is also possible to determine landslide effects by comparing DEMs of before and after the landslide event.
- **Automatic catchment delineation:**
Catchment boundaries or drainage lines can be automatically generated from a good quality DEM with the use of neighbourhood functions. The system will determine the lowest point in the DEM, which is considered the outlet of the catchment. From there, it will repeatedly search the neighbouring pixels with the highest altitude. This process is continued until the highest location (i.e. cell with highest value) is found, and the path followed determines the catchment boundary. For delineating

the drainage network, the process is reversed. Now, the system will work from the watershed downwards, each time looking for the lowest neighbouring cells, which determines the direction of water flow.

- **Dynamic modelling:**

Apart from the applications mentioned above, DEMs are increasingly used in GIS-based dynamic modelling, such as the computation of surface run-off and erosion, groundwater flow, the delineation of areas affected by pollution, the computation of areas that will be covered by processes such as debris flows and lava flows.

- **Visibility analysis:**

A viewshed is the area that can be 'seen'—i.e. is in the direct line-of-sight—from a specified target location. Visibility analysis determines the area visible from a scenic lookout, the area that can be reached by a radar antenna, or assesses how effectively a road or quarry will be hidden from view.



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6. Perform the raster overlay operation to find R₃ (APR 19)

$$R_3 = \text{CON}(R_1=3 \text{ AND } (R_2 \geq 45 \text{ and } R_2 \leq 60), 1, 0)$$

R₁ - Soil Type Raster

1	1	1	3	3	4	4	7	7
1	1	3	3	4	4	4	7	7
1	3	3	3	4	4	4	7	7
3	3	3	3	4	4	4	5	5
5	5	3	3	3	3	3	5	5
2	2	3	1	1	1	3	4	4
2	2	3	1	1	1	3	4	4
7	2	3	1	1	1	3	4	4
7	7	3	3	3	3	3	4	4

R₂ - Rainfall Raster (mm)

70	70	70	70	60	60	60	60	50
70	70	70	70	80	80	80	50	50
50	50	50	80	80	80	50	50	50
50	50	90	90	90	60	60	50	50
50	35	35	35	50	45	60	60	70
35	35	35	50	50	45	60	60	70
45	35	35	50	45	45	60	60	60
45	35	45	45	45	60	60	70	70
45	45	45	45	60	60	70	70	70

ANS:-

R₃ – Soil Type Raster

0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0
0	0	0	0	0	1	1	0	0
0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	1	0	0
0	0	1	0	0	0	1	0	0
0	0	1	1	1	1	1	0	0

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7. Write a note on neighborhood functions.**(Nov 19)(Apr 23)****Ans****❖ Neighbourhood functions:**

There is another guiding principle in spatial analysis that can be equally useful. The principle here is to find out the characteristics of the vicinity, here called neighbourhood, of a location.

After all, many suitability questions, for instance, depend not only on what is at the location, but also on what is near the location.

Thus, the GIS must allow us ‘to look around locally’. To perform neighbourhood analysis, we must:

1. State which target locations are of interest to us, and define their spatial extent,
2. Define how to determine the neighbourhood for each target,
3. Define which characteristic(s) must be computed for each neighbourhood. Then, in the third step we indicate what it is we want to discover about the phenomena that exist or occur in the neighbourhood. This might simply be its spatial extent, but it might also be statistical information like:
 - The total population of the area,
 - Average household income, or
 - The distribution of high-risk industries located in the neighbourhood.

The above are typical questions in an urban setting. When our interest is more in natural phenomena, different examples of locations, neighborhoods and neighborhood characteristics arise. Since raster data are the more commonly used in this case, neighborhood characteristics often are obtained via statistical data on the cells compute values such as average, minimum, maximum, and standard deviation of the cells in the identified neighborhood

➤ Determining neighbourhood extent:

To select target locations, one can use the selection techniques. To obtain characteristics from an eventually identified neighbourhood, the same techniques apply. So what remains to be discussed here is the proper determination of a neighbourhood.



8. What is Classification of data in GIS? Explain using suitable example.

(Nov 19)

Ans:

Classification is a technique of purposefully removing detail from an input dataset, in the hope of revealing important patterns (of spatial distribution). In the process, we produce an output data set, so that the input set can be left intact.

➤ User-controlled classification:

In user-controlled classification, a user selects the attribute(s) that will be used as the classification parameter(s) and defines the classification method. The latter involves declaring the number of classes as well as the correspondence between the old attribute values and the new classes. This is usually done via a classification table.

➤ Automatic classification:

User-controlled classifications require a classification table or user interaction. GIS software can also perform automatic classification, in which a user only specifies the number of classes in the output data set. The system automatically determines the class break points. Two main techniques of determining break points are in use.

1. Equal interval technique: The minimum and maximum values v_{min} and v_{max} of the classification parameter are determined and the (constant) interval size for each category is calculated as $(v_{max} - v_{min})/n$, where n is the number of classes chosen by the user.
2. Equal frequency technique: This technique is also known as quantile classification. The objective is to create categories with roughly equal numbers of features per category.

1	1	1	2	8
4	4	5	4	9
4	3	3	2	10
4	5	6	8	8
4	2	1	1	1

(a) original raster

1	1	1	1	4
2	2	3	2	5
2	2	2	1	5
2	3	3	4	4
2	1	1	1	1

(b) equal interval classification

1	1	1	2	5
3	3	4	3	5
3	2	2	2	5
3	4	4	5	5
3	2	1	1	1

(c) equal frequency classification

original value	new value	# cells
1,2	1	9
3,4	2	8
5,6	3	3
7,8	4	3
9,10	5	2

original value	new value	# cells
1	1	6
2,3	2	5
4	3	6
5,6	4	3
8,9,10	5	5

Figure 6.11: Example of two automatic classification techniques: (a) the original raster with cell values; (b) classification based on equal intervals; (c) classification based on equal frequencies. Below, the respective classification tables, with a tally of the number of cells involved.

9. Write a note on GIS and application models.

(Nov 19)

Ans:

GIS and application models:

We define application models to include any kind of GIS based model (including so-called analytical and process models) for a specific real-world application. Such a model, in one way or other, describes as faithfully as possible how the relevant geographic phenomena behave, and it does so in terms of the parameters. The nature of application models varies enormously. GIS applications for famine relief programs, for instance, are very different from earthquake.

Risk assessment applications, though both can make use of GIS to derive a solution. Many kinds of application models exist, and they can be classified in many different ways. Here we identify five characteristics of GIS-based application models:

1. The purpose of the model,
2. The methodology underlying the model,
3. The scale at which the model works,
4. Its dimensionality- i.e. whether the model includes spatial, temporal or spatial and temporal dimensions, and
5. Its implementation logic- i.e. the extent to which the model uses existing knowledge about the implementation context.

It is important to note that the categories above are merely different characteristics of any given application model. Any model can be described according to these characteristics. Each is briefly discussed below.

Purpose of the model refers to whether the model is descriptive, prescriptive or predictive in nature. Descriptive models attempt to answer the "what is" question. Prescriptive models usually answer the "what should be" question by determining the best solution from a given set of conditions.

Models for planning and site selection are usually prescriptive, in that they quantify environmental, economic and social factors to determine 'best' or optimal locations. So-called Predictive models focus upon the "what is likely to be" questions, and predict outcomes based upon a set of input conditions. Examples of predictive models include forecasting models, such as those attempting to predict landslides or sea-level rise.

Methodology refers to the operational components of the model. Stochastic models use statistical or probability functions to represent random or semi-random behaviour of phenomena: In contrast, deterministic models are based upon a well-defined cause and effect relationship. Examples of deterministic models include hydrological flow and pollution models, where the 'effect can often be described by numerical methods and differential equations. Rule-based models attempt to model processes by using local (spatial) rules. Cellular Automata (CA) are examples of models in this category. These are often used to understand systems which are generally not well understood, but for which their local processes are well known. For example, the characteristics of neighborhood cell to model wind direction and vegetation type, the characteristics of fire used to the direction of spread of a fire over several time steps.

10. How error propagates in data processing? Explain.

(Nov 19)

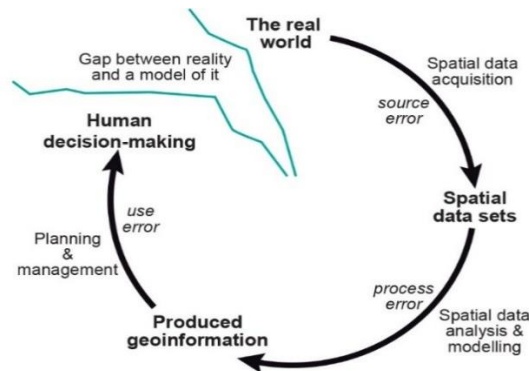
OR

How Error Propagates in GIS? Explain using suitable diagram. (Apr 23)

Ans:

➤ **Error propagation in spatial data processing:**

A number of sources of error that may be present in source data. It is important to note that the acquisition of base data to a high standard of quality still does not guarantee that the results of further, complex processing can be treated with certainty. As the number of processing steps increases, it becomes difficult to predict the behaviour of error propagation. These various errors may affect the outcome of spatial data manipulations. In addition, further errors may be introduced during the various processing steps discussed earlier in this chapter, as illustrated in Figure.



One of the most commonly applied operations in geographic information systems is analysis by overlaying two or more spatial data layers. As discussed above, each such layer will contain errors, due to both inherent inaccuracies the source data and errors arising from some form of computer processing, for example, rasterization.

During the process of spatial overlay, all the errors in the individual data layers contribute to the final error of the output. The amount of error in the output depends on the type of overlay operation applied. For example, errors in the results of overlay using the logical operator AND are not the same as those created using the OR operator.

Table contains lists common sources of error introduced into GIS analyses. Note that these are from a wide range of sources, and include various common tasks relating to both data preparation and data analysis. It is the combination of different errors that are generated at each stage of preparation and analysis which may bring about various errors and uncertainties in the eventual outputs.

Consider another example. A land use planning agency is faced with the problem of identifying areas of agricultural land that are highly susceptible to erosion. Such areas occur on steep slopes in areas of high rainfall. The spatial data used in a GIS to obtain this information might include:

- A land use map produced five years previously from 1 : 25, 000 scale aerial photographs,
- A DEM produced by interpolating contours from a 1 : 50, 000 scale topographic map, and Annual rainfall statistics collected at two rainfall gauges.

11. Perform the raster overlay operation to find $R_4 = R_1 \text{ AND } (R_2 \text{ OR } R_3)$
(Nov 19)

R1					R2					R3				
1	1	0	1		1	0	0	0		1	1	1	1	
0	1	0	0		1	1	0	0		0	1	1	1	
0	0	1	1		1	1	1	0		0	0	1	1	
1	0	1	0		1	1	1	1		0	0	0	1	

Ans:

Raster Overlay Operation on $R_4 = R_1 \text{ AND } (R_2 \text{ OR } R_3)$

R2 OR R3			
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

R4=R1 and (R2 OR R3)			
1	1	0	1
0	1	0	0
0	0	1	1
1	0	1	0

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12. Explain the two main techniques of determining Automatic classification.

(Apr 23)

Ans:

➤ Automatic Classification

User-controlled classifications require a classification table or user interaction. GIS software can also perform automatic classification, in which a user only specifies the number of classes in the output data set. The system automatically determines the class break points. Two main techniques of determining break points are in use.

1. Equal interval technique

The minimum and maximum values v_{min} and v_{max} of the classification parameter are determined and the (constant) interval size for each category is calculated as $(v_{max} - v_{min})/n$, where n is the number of classes chosen by the user. This classification is useful in revealing the distribution patterns as it determines the number of features in each category.

2. Equal frequency technique

This technique is also known as quantile classification. The objective is to create categories with roughly equal numbers of features per category. The total number of features is determined first and by the required number of categories, the number of features per category is calculated. The class break points are then determined by counting off the features in order of classification parameter value.



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13. Write a short note on Network Analysis.**(Apr 23)****Ans:**

A completely different set of analytical functions in GIS consists of computations on networks. A network is a connected set of lines, representing some geographic phenomenon, typically of the transportation type. The 'goods' transported can be almost anything: people, cars and other vehicles along a road network, commercial goods along a logistic network, phone calls along a telephone network, or water pollution along stream/river network.

Network analysis can be performed on either raster or vector data layers, but they are more commonly done in the latter, as line features can be associated with a network, and hence can be assigned typical transportation characteristics such as capacity and cost per unit. A fundamental characteristic of any network is whether the network lines are considered directed or not.

Associate with each line a direction of transportation; undirected networks do not. The 'goods' can be transported along a line in both directions.

For many applications of network analysis, a planar network, ie, one that can be embedded in a two-dimensional plane, will do the job. Many networks are naturally planar, like stream/river networks. A large-scale traffic network, on the other end, is not planar: motorways have multi-level crossings and are constructed with underpasses and overpasses. Planar networks are easier to deal with computationally, as they have simpler topological rules.

Not all GISs accommodate non-planar networks, or can do so only using 'tricks these may involve the splitting of overpassing lines at the intersection vertex and the creation of four lines out of the two original lines. Without further attention, the network will then allow one to make a turn onto another line at this new intersection node, which in reality would be impossible. In some GISs we can allocate a cost with turning at a node and that cost, in the case of the overpass, can be made infinite to ensure it is prohibited. But, as mentioned, this is a workaround to fit a non-planar situation into a data layer that presumes planarity. This is a good illustration of geometry not fully determining the network's behaviour.

Additional application-specific rules are usually required to define what can and cannot happen in the network. Most GISs provide rule-based tools that allow the definition of these extra application rules. Various classical spatial analysis functions on networks are supported by GIS software packages.

14. Perform the raster overlay operation to project Ground Water Level Raster in 2025

$$R2 = \text{con}(R1 > 5, R1 - 5, 0)$$

R1- Ground Water Raster in 2023

7	8	8	4	3	3
6	4	4	5	4	4
7	10	10	8	7	4
4	8	8	8	7	4
1	1	1	3	0	0
1	0	0	7	0	0

(APR 23)

Ans:

2	3	3	0	0	0
1	0	0	0	0	0
2	5	5	3	2	0
0	3	3	3	2	0
0	0	0	0	0	0
0	0	0	2	0	0

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15.What is overlay functions?

Ans:

Overlay is a technique of combining two spatial data layers and producing a third from them. The binary operators that we discuss are known as spatial overlay operators. We will firstly discuss vector overlay operators, and then focus on the raster case. Standard overlay operators take two input data layers, and assume they are georeferenced in the same system, and overlap in study area. If either of these requirements is not met the use of an overlay operator is senseless. The principle of spatial overlay is to compare the characteristics of the same location in both data layers, and to produce a result for each location in the output data layer. The specific result to produce is determined by the user. It might involve a calculation, or some other logical function to be applied to every area or location.

In raster data, as we shall see, these comparisons are carried out between pairs of cells, one from each input raster. In vector data, the same principle of comparing locations applies, but the underlying computations rely on determining the spatial intersections of features from each input layer.

➤ Vector Overlay Operators

In the vector domain, overlay is computationally more demanding than in the raster domain. Here we will only discuss overlays from polygon data layers, but we note that most of the ideas also apply to overlay operations with point or line data layers.

The standard overlay operator for two layers of polygons is the polygon intersection operator. It is fundamental, as many other overlay operators proposed in the literature or implemented in systems can be defined in terms of it. The result of this operator is the collection of all possible polygon intersections; the attribute table result is a join-in the relational database of the two input attribute tables. This output attribute table only contains one table for each intersection polygon found, and this explains why we call this operator a spatial join.

➤ Raster Overlay Operators

Vector overlay operators are useful, but geometrically complicated, and this sometimes results in poor operator performance. Raster overlays do not suffer from this disadvantage, as most of them perform their computations cell by cell, and thus they are fast. GIS that support raster processing have a language to express operations on raster referred to as map algebra, or raster calculus, allowing a GIS to compute new raster from existing ones, using a range of functions and operators.

The key operations using a logical structured language differs for different GIS software packages. When producing a new raster we must provide a name for it, and define how it is computed.

16. Explain spatial queries?

Ans:

Spatial queries in GIS (Geographic Information Systems) involve retrieving, analyzing, and manipulating geographic data based on their spatial relationships. These queries are essential for extracting meaningful information from spatial datasets and gaining insights into the geographic characteristics of the data.

When exploring a spatial data set, the first thing one usually wants is to select certain features, to (temporarily) restrict the exploration. Such selections can be made on geometric/spatial grounds, or on the basis of attribute data associated with the spatial features.

- **Interactive spatial selection:**

In interactive spatial selection, one defines the selection condition by pointing at or drawing spatial objects on the screen display, after having indicated the spatial data layer(s) from which to select features. The interactively defined objects are called the selection objects; they can be points, lines, or polygons. The GIS then selects the features in the indicated data layer(s) that overlap i.e. intersect, meet, contain, or are contained in; as show in Figure below with the selection objects.

Interactive spatial selection answers questions like "What is at...?" In Figure above, the selection object is a circle and the selected objects are the black and orange dots; they overlap with the selection object. All city regions that overlap with the selection object indicated using dots, and their corresponding attribute records are highlighted in the data table.

- **Spatial selection by attribute conditions:**

It is also possible to select features by using selection conditions on feature attributes. These conditions are formulated in SQL if the attribute data reside in a geodatabase.

- **Combining attribute conditions:**

When multiple criteria have to be used for selection, we need to carefully express all of these in a single composite condition. The tools for this come from a field of mathematical logic, known as propositional calculus.

Spatial selection using topological relationships:

Various forms of topological relationship can be useful to select features as well. The steps carried out are:

1. To select one or more features as the selection objects, and
2. To apply a chosen spatial relationship function to determine the selected features that have that relationship with the selection objects.

17. What are different GIS models?

Ans:

GIS (Geographic Information System) models are representations of the real world that help organize and analyze geographic data. These models serve as frameworks for understanding spatial relationships and patterns. Several GIS models exist, each designed to address specific aspects of spatial analysis and representation. Here are some of the different GIS models:

➤ **Vector Data Model:**

The vector data model represents geographic features as points, lines, and polygons. Points are used to represent discrete locations (e.g., cities), lines represent linear features (e.g., roads), and polygons represent areas (e.g., land parcels). Attributes are associated with each feature, forming a table linked to the spatial data.

➤ **Raster Data Model:**

The raster data model represents geographic features as a grid of cells, where each cell has a value representing a specific attribute. It is suitable for continuous data, such as elevation models, temperature distributions, and land cover. Raster data is organized in rows and columns, forming a regular grid.

➤ **Triangular Irregular Network (TIN) Model:**

The TIN model represents terrain surfaces using a network of non-overlapping triangles. It is particularly useful for modeling irregular terrain with varying elevation. TINs provide a more accurate representation of terrain morphology compared to regular grid-based models.

➤ **Geodatabase Model:**

The geodatabase model is an integrated data model that incorporates both spatial and attribute data. It provides a comprehensive framework for organizing and managing geographic information in a relational database, offering more advanced capabilities than traditional file-based approaches.

➤ **Network Data Model:**

The network data model is designed for representing and analyzing network features such as road networks, utility networks, and telecommunication networks. It includes nodes and edges, where nodes represent junctions, and edges represent the connections between nodes.

➤ **Hybrid Data Model:**

The hybrid data model combines elements of both the vector and raster models, allowing users to represent and analyze spatial features in a more flexible way. This model is advantageous when dealing with complex geographic phenomena that require both discrete and continuous representations.

➤ **Object-Oriented Data Model:**

The object-oriented data model represents geographic entities as objects with associated attributes and behaviors. This model is useful for managing complex spatial relationships and supporting more advanced analysis in GIS.

➤ **Semantic Data Model:**

The semantic data model focuses on capturing the meaning and relationships between geographic features. It allows for the representation of complex spatial semantics and supports richer spatial analysis by considering the semantics of spatial entities.

➤ **Cellular Automata Model:**

Cellular automata are used to model dynamic spatial processes and changes over time. It divides space into cells, and each cell's state evolves based on predefined rules and the states of neighboring cells. This model is often used for simulating landscape changes and urban growth.

➤ **Agent-Based Model (ABM):**

ABM represents entities (agents) that interact with each other and their environment. Agents have individual behaviors and can adapt to changing conditions. This model is used to simulate complex systems, such as transportation patterns, land-use changes, and ecosystem dynamics.

These GIS models provide different ways of representing and analyzing geographic information, and the choice of model depends on the nature of the data and the specific requirements of the analysis. Integrating these models allows for more comprehensive and accurate representations of the real world in GIS applications.



18. What is quantifying error propagation?

Ans:

Chrisman noted that “the ultimate arbiter of cartographic error is the real world, not a mathematical formulation”. It is an unavoidable fact that we will never be able to capture and represent everything that happens in the real world perfectly in a GIS. Hence there is much to recommend the use of testing procedures for accuracy assessment.

Various perspectives, motives and approaches to dealing with uncertainty have given rise to a wide range of conceptual models and indices for the description and measurement of error in spatial data. All these approaches have their origins in academic research and have strong theoretical bases in mathematics and statistics. Here we identify two main approaches for assessing the nature and amount of error propagation:

- 1 Testing the accuracy of each state by measurement against the real world, and
- 2 Modelling error propagation, either analytically or by means of simulation techniques.

Modelling of error propagation has been defined by Veregin as: “the application of formal mathematical models that describe the mechanisms whereby errors in source data layers are modified by particular data transformation operations.” In other words, we would like to know how errors in the source data behave under manipulations that we subject them to in a GIS. If we are able to quantify the error in the source data as well as their behaviour under GIS manipulations, we have a means of judging the uncertainty of the results.

Error propagation models are very complex and valid only for certain data types (e.g. numerical attributes). Initially, they described only the propagation of attribute error. More recent research has addressed the spatial aspects of error propagation and the development of models incorporating both attribute and locational components. These topics are outside the scope of this book, and readers are referred to for more detailed discussions. Rather than explicitly modelling error propagation, it is often more practical to test the results of each step in the process against some independently measured reference data.

Looking at this robust system, its applications and uses are never-ending, just like its vast amount of geospatial data sets and databases. Day by day, analysts and researchers are innovating new applications of this technology. The count of the applications is never going to fall. Using GIS is not just limited to above 6 -7 applications, but it has around 1000+ uses and applications in various fields.

Archaeology, geology, Waste management, Natural Resources Management, Asset management and even Aviation and Banking. It seems that in the near future, GIS is going to get integrated with everything, and that's why companies like HERE Maps and GOOGLE are working on prototypic concepts like the Internet of Things, where everything will be interconnected.



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